CE-04: MicroMacro Error Under SIR Dynamics

Experiment Report — Two-Scale CTMC Project

Claude Experiments Pipeline

July 31, 2026

Abstract

This report examines how MicroMacro approximation error scales with the recovery rate γ in an SIR epidemic model. All scenarios share $\beta = 0.003$, chain-of-cliques topology ($L = 10$, $n = 100$, $d = 6$, $k = 5$), and 100 paired replications. The error ΔRMST (MicroMacro − full Gillespie) is significantly negative across all scenarios and grows monotonically with γ. Recovery exacerbates premature hazard export because the macro layer does not account for wave passage: boundary nodes may have already recovered by the time the exported macro hazard fires. *Ukrainian summary*: see `docs/ce04_sir_validation/report.md`.

1 Experimental Design

1.1 Objective

Determine whether adding SIR recovery ($\gamma > 0$) to the SI baseline amplifies or reduces MicroMacro approximation error.

1.2 Network and Parameters

- Topology: chain-of-cliques, $L = 10$ communities, $n = 100$ nodes each, degree $d = 6$, $k = 5$ inter-community edges.
- Infection rate: $\beta = 0.003$ (within and between communities).
- Recovery rates: $\gamma \in \{0,\ 0.0005,\ 0.001,\ 0.003\}$.
- Basic reproduction number: $R_0 = \beta/\gamma \in \{\infty, 6, 3, 1\}$.
- Replications: 100 paired runs (same seed for Micro and MicroMacro).
- Horizon: $T = 5000$ time units; epidemic declared complete when no infected nodes remain or $t > T$.

1.3 Primary Metric

$$\Delta\text{RMST} = \overline{T_{\text{MM}}} - \overline{T_{\text{Micro}}}$$

where T denotes mean first arrival time at the last community. A negative ΔRMST indicates premature arrival under MicroMacro. Bootstrap 95% confidence intervals use 10,000 resamples.

2 Results

Table 1: CE-04 results: MicroMacro error by recovery rate. ΔRMST = mean difference (MicroMacro − Micro). Significant if CI excludes zero.

Scenario	γ	R_0	ΔRMST	95% CI	Sig.
SI	0.0000	∞	−1067.6	[−1242.5, −917.7]	*
SIR weak	0.0005	6.0	−1244.3	[−1428.3, −1054.1]	*
SIR mid	0.0010	3.0	−1587.3	[−1788.0, −1375.2]	*
SIR strong	0.0030	1.0	−1726.3	[−2020.4, −1426.1]	*

All four scenarios are statistically significant. The error grows monotonically with γ (rows 1–4 are final).

2.1 Relative Wasserstein Distance

Relative W_1 distance and epidemic completion rates:

Table 2: Completion rates and distributional error by scenario. *Completion* = fraction of 100 replications in which the epidemic reached the last community.

Scenario	γ	Micro compl.	MM compl.	rel-W_1
SI	0.0000	100%	100%	0.3706
SIR weak	0.0005	89%	96%	0.4102
SIR mid	0.0010	70%	93%	0.4703
SIR strong	0.0030	28%	72%	0.4399

Critical observation: At $\gamma = 0.003$ ($R_0 = 1$), only 28% of full Gillespie replications produce a chain-wide epidemic, but MicroMacro declares completion in 72% of replications — a **false-positive rate of 44 percentage points**. This is a qualitatively different failure mode: MicroMacro does not merely predict early arrival, it predicts *arrival when no arrival would have occurred* in the stochastic reality.

3 Interpretation

3.1 Mechanism of Error Amplification

Under the two-scale approximation, the macro layer fires an import event using the aggregate fraction $\bar{i}(t)$ of infected nodes at the moment of export. In an SIR model, $\bar{i}(t)$ first rises (infection wave) and then falls (recovery). There are two failure modes:

1. **Wave-passage mismatch.** After the peak, $\bar{i}(t)$ is falling. If a macro transition fires when the wave has already passed, the hazard is smaller than it was at the wave's peak, yet the event fires because the aggregate macro rate is integrated over the full wave shape, not the current one. Result: artificially early arrivals.

2. **Censoring effect.** With high γ, many replications reach $R_0 \approx 1$ and fail to produce a full chain-wide epidemic. These censored runs (no arrival) are absent from the MicroMacro distribution but inflate the Micro mean — or are absent from both, depending on implementation. The *conditional* mean under MicroMacro is then far more negative.

3.2 Significance for the Paper

The SI case is the paper's primary scenario. The CE-04 results show:

- MicroMacro is premature even in the SI baseline ($\Delta\text{RMST} \approx -1068$).
- Recovery ($\gamma > 0$) monotonically *worsens* this error.
- The paper's claim about premature hazard export is thus conservative: the real-world SIR error is larger than the SI error reported.
- Near $R_0 = 1$ ($\gamma = 0.003$), MicroMacro generates a 44-percentage-point false-positive rate for epidemic completion. This is a qualitatively distinct failure: the model predicts epidemics that would *never occur* in a stochastic simulation.
- The rel-W_1 distance is non-monotone at $\gamma = 0.003$ (drops from 0.4703 to 0.4399) because most censored Micro runs have no arrival time, compressing the reference distribution.

4 Conclusion

Recovery amplifies MicroMacro approximation error in two distinct ways. First, the conditional ΔRMST grows monotonically: from -1068 (SI) to -1726 (SIR, $\gamma = 0.003$), a 62% increase. Second, and more critically, near the epidemic threshold ($R_0 = 1$), MicroMacro generates 44 percentage points more false epidemic completions than the full stochastic model. Both failure modes trace to the wave-passage mismatch: the aggregate macro hazard does not account for the post-peak decline of $\bar{i}(t)$ nor for the probability that individual boundary nodes have already recovered. Future work should develop bridge-aware hazard corrections that condition on gateway-node infection state, not on the aggregate community fraction.

Data Location

`output/claude_experiments/ce04_sir_validation/`